MIL-F-48173 (MU)
15 May 1973
SUPERSEDING
PA-PD-3055
22 NOVEMBER 1967

MILITARY SPECIFICATION

FUZE, M814, TACTICAL AND TELEMETRY PARTS, ASSEMBLING, LOADING AND PACKING FOR

1. SCOPE

1.1 This specification contains requirements not normally required by the drawings and provides the quality assurance provisions for the fabrication of parts, assembly, loading and packing of one type of fuze designated as Fuze, M814, Tactical and Telemetry.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550 - Ammunition and Special Weapons, General Specification for
MIL-D-48221 - Detonator, Electric, M95, Parts for, Loading, Assembling and Packing
MIL-S-60902 - Soldering Electrical Connections and Printed Wiring Assemblies, Procedures for
MIL-I-45607 - Inspection Equipment, Supply and Maintenance of

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109 - Quality Assurance Terms and Definitions

FSC: 1340

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MIL-STD-331 - Fuze and Fuze Components, Environmental and Performance Tests for

Test No. 101
Test No. 102
Test No. 103

MIL-STD-1167 - Ammunition Data Cards
MIL-STD-1168 - Lot Numbering of Ammunition
MIL-STD-1169 - Packaging, Packing and Marking for Shipment of Inert Ammunition Components
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

US ARMY MUNITIONS COMMAND

P-9227896 - M814 Fuze Assembly
P-9227902 - TM M814 Fuze Assembly

(Copies of specifications, standards, drawings and publications required by the supplier in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer.)

2.2 Other Publications. - The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

CODE OF FEDERAL REGULATIONS

49 CFR 0-199 - Department of Transportation Rules for Regulations for the Transportation of Explosive and Other Dangerous Articles

(The Department of Transportation Regulations are now a part of the Code of Federal Regulations [1949 Edition and revisions] available from the Superintendent of Documents, Government Printing Office, Washington, D.C. Orders for the above publication should cite "49 CFR 0-199 [latest revisions]".)

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3. REQUIREMENTS

3.1 Material. - Material shall be in accordance with applicable drawings and all referenced specifications.

3.2 Parts. - The parts shall comply with all requirements specified in the applicable drawings and specifications.

3.3 Performance. - The fuze shall perform as follows when subjected to conditions specified in 3.5 (environments).

3.3.1 Fuze Activation

3.3.1.1 Interlock (MMS 1.0 Switch). - The unlock function (MMS 1.0 activation) of the fuze shall occur when supplied with 17 volts minimum to 45 volts maximum with a minimum internal resistance of 5.64 ohms for 0.001 seconds maximum. The interlock function shall not occur when supplied with a maximum of 0.05 amps for 30 seconds minimum. The interlock resistance shall be 7 $\pm$ 2 ohms when tested at 0.005 amps maximum for 30 seconds maximum.

3.3.1.2 Boost Switch. - The boost switch shall function when subjected to a longitudinal acceleration (M-N axis of Figure 1) of 33 g's for 0.006 seconds minimum and shall not function below 23 g's. The attainment of 33 g's shall be referred to as zero time.

3.3.2 Sustainer Motor Ignition. - Prior to sustainer motor ignition time, the sustainer ignition lead shall be electrically connected to fuze ground. At 0.310 $\pm$ 0.045 seconds from zero time, the sustainer ignition lead shall be electrically transferred from fuze ground to fuze B plus (B+) for a period of 0.200 $\pm$ 0.100 seconds. After the 0.200 $\pm$ 0.100 second period, the sustainer ignition lead shall be returned to fuze ground.

3.3.3 Fuze Arming. - The fuze shall arm at 1.75 $\pm$ 0.30 seconds after zero time when subjected to a longitudinal acceleration of 33 g's. The fuze shall not arm when subjected to a longitudinal acceleration below 5 g's and shall arm when subjected to a longitudinal acceleration of 10 g's.

3.3.4 Penetration Impulse Generator (P.I.G). - The explosive train of the fuze shall fire high order when a .040 inch thick aluminum target of 2.25 $\pm$ .010 inch inside diameter is fired over the PIG at a speed of 500 feet per second maximum.

3.3.5 Impact Switch (IMP). - The impact switch shall function upon sensing a longitudinal deceleration of 33 g's or more for .006 seconds minimum. The impact switch shall not function as a result of a longitudinal deceleration of less than 23 g's.

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3.3.6 Self Destruct (S.D). - Self destruct shall occur at 13.5 to 20 seconds from zero time if the fuze is armed.

3.3.7 Piston Actuator Resistance. - The piston actuator must have a resistance of 2.0 ± 0.6 ohms.

3.3.8 M95 Detonator. - The detonators shall meet the requirements of MIL-D-48221 with the following exceptions:

a. Paragraph 4.2 shall be deleted.

b. Paragraph 4.3.3.6.2, Part c shall be deleted and the following substituted: "Duds. - The lot shall be rejected if two (2) or more detonators fail the time, output or dud test. If no failures occur, the lot shall be acceptable. If one failure occurs out of the 185 sample, a second sample of 185 shall be randomly selected from the lot and subjected to the functioning test of Paragraph 4.3.3.6.1 of MIL-D-48221. If no failures occur on the second sample, the lot is acceptable. If one or more failure occurs on the second sample, the lot is rejected."

c. Paragraph 4.4.7.3 shall be deleted and the following substituted:

"4.4.7.3 Half the detonators shall be conditioned at -45°F and half at +140°F for a minimum of 24 hours. Any odd unit shall be conditioned at -45°F. Detonator assemblies shall then be assembled in the test fixture specified in 4.3.4 and fired by applying the same energy specified in 4.4.7.1 within a maximum of four (4) minutes from the time the detonator is removed from the conditioning chamber. Observation shall be made to determine compliance with the requirements."

3.3.9 M95 Detonator. - The resistance of the detonator shall be between 1,000 and 10,000 ohms when measured prior to installation in the fuze mechanism.

3.4 Complete Fuze Circuit Continuity. - The complete fuze shall exhibit the following properties when the circuitry is measured for electrical continuity between the following points (see dwg. P-9227896 for identification of points):

Measurement Points	Property Measured
J3 Pin 1/J3 Pin 5	0.2 ohm max.
J3 Pin 3/J3 Pin 5	0.2 ohm max.
J3 Pin 1/J3 Pin 2	7 ± 2 ohm
J2 Pin 1/J3 Pin 1	0.2 ohm max.

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Measurement Points	Property Measured
J2 Pin 5/J3 Pin 2	0.2 ohm max.
J2 Pin 4/J3 Pin 2	1 megohm min.
J2 Pin 3/J2 Pin 5	Capacitor charge to 1 megohm min.
J2 Pin 1/Case	50 megohms min.
J2 Pin 4/J3 Pin 3	1 megohm min.
J2 Pin 3/J4 Pin 2	5 ohms max.
J4 Pin 1/J4 Pin 2	5 ohms max.
J2 Pin 3/J4 Pin 1	5 ohms max.
J2 Pin 1/J2 Pin 4	Capacitor Charge
J2 Pin 4/J2 Pin 3	15K ± 10K ohm

3.5 Environmental Requirements. - The fuze shall meet the performance requirements (see 3.3) of this specification after being subjected to the following:

3.5.1 Vibration, Field Service -

a. Intensity:	10 ± 1 gravity units (g's)
b. Frequency:	50 ± 2.5 cycles per second (cps)
c. Duration:	60 ± 3 minutes
d. Application:	Parallel to each axis of Figure 1, applied to entire outer surface (no cantilevering).

3.5.2 Handling Shock -

a. Intensity:	400 ± 32 g's
b. Rise Time:	0.002 + 0.0000, - 0.0005 seconds
c. Decay Time:	0.002 + 0.0000, - 0.0005 seconds
d. Direction:	Perpendicular to M-N axis of Figure 1 (any quadrant).
e. Application:	Entire outer surface (no cantilevering)

3.5.3 High Temperature, Operative - +60° ± 3°C (+140° ± 5°F).

3.5.4 Low Temperature, Operative - -43° ± 3°C (-45° ± 5°F).

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3.5.5 Vibration-Acceleration, Operative -

a. Longitudinal:

(1)	Acceleration and Frequency:	14 g's peak, at 30 ± 1 cps
(2)	Superimposed Acceleration:	5 g's peak, with an exponential sweep from 40 to 2000 cps
(3)	Duration:	25 ± 5 seconds

b. A-Plane (Parallel to the P-R Axis of Figure 1):

(1)	Acceleration and Frequency:	24 g's peak, at 30 ± 1 cps
(2)	Superimposed Acceleration:	5 g's peak, with an exponential sweep from 40 to 2000 cps
(3)	Duration:	25 ± 5 seconds

c. B-Plane (Parallel to S-T Axis of Figure 1):

(1)	Acceleration and Frequency:	16 g's peak, at 30 ± 1 cps
(2)	Superimposed Acceleration:	5 g's peak, with an exponential sweep from 40 to 2000 cps
(3)	Duration:	25 ± 5 seconds

3.5.6 Spin -

a.	Constant:	12 ± 2 R.P.S.
b.	Direction:	Counterclockwise looking forward along the axis of Figure 1

3.6 Safety Requirements. - The piston actuator and the 5 to 10 g lock independently shall keep the fuze from arming when subjected to the tests below. The fuze shall also meet all other requirements of the tests below.

a. Jolt - Per MIL-STD-331, Test No. 101
b. Jumble - Per MIL-STD-331, Test No. 102
c. 40 Foot Drop - Per MIL-STD-331, Test No. 103

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3.7 Soldering Electrical Connections. - All electrical connections shall conform to MIL-S-60902.

3.8 Workmanship. - The fuze shall be fabricated and finished in a thorough, workmanlike manner. It should be free of burrs, chips, sharp edges, cracks, unblended radii, surface defects, dirt, grease, rust, corrosion products and other foreign matter. The cleaning methods used shall not be injurious to any part, nor shall the parts be contaminated by the cleaning agents.

3.9 Spring Embrittlement. - Spring components shall not break when extended or distorted to the point of minimum permanent distortion.

3.10 First Article Inspection. - This specification makes provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 in order to define terms used herein. The provisions of MIL-A-2550 shall apply.

4.1.1 Submission of Product. - At the time each completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified in this specification.

b. Drawing and specification number and date, together with identification and date of changes thereto.

c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review.

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d. Number of items in the lot.
e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required by subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 First Article Inspection

4.2.1 Submission. - The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.2.2. The first article sample shall consist of five (5) complete sets of parts, sixteen (16) M814 Fuze Assemblies (P-9227896) and five (5) sets of the following assemblies:

a. (P-9227795) - Boost Switch Plug Assembly
b. (P-9227805) - Pallet Assembly
c. (P-9227813) - Escapement Wheel and Pinion Assembly
d. (P-9227839) - Lead Cup Assembly
e. (P-9227843) - Hookup Wire (Black)
f. (P-9227844) - Hookup Wire (Red)
g. (P-9227845) - Hookup Wire (Yellow)
h. (P-9227846) - Hookup Wire (Blue)
i. (P-9227870) - Rotor Lead Assembly
j. (P-9227889) - Mechanism Assembly (less Detonator and Piston Actuator
k. (P-9227889) - Mechanism Assembly
l. (P-9227890) - Body Assembly
m. (P-9227891) - Timer Assembly
n. (P-9227893) - Printed Wiring Board Assembly

which have been produced by the contractor or furnished by a supplier and which have been manufactured using the same production processes, procedures and equipment which will be used in fulfilling the contract. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as will be used in regular production. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample, including statements of findings of tests and examinations and certificates of conformance for materials (see 4.1.1.c). A sample containing known defects will not be submitted unless specifically authorized by the Contracting Officer. A first article sample, or portion thereof as directed by the Contracting Officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days

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or whenever a change occurs in manufacturing process, material used, drawing, specification or source of supply such as to significantly affect product uniformity as determined by the Government.

4.2.2 Inspections to be Performed. - Each assembly in the first article sample shall be examined and inspected for every dimension and requirement of the applicable drawings. The entire sample shall be examined in accordance with 4.3.2 and then tested as specified in 4.3.3.

4.2.3 Rejection. - The first article sample shall be rejected if one or more fuzes, subassemblies or components fail to conform to any or all of the applicable requirements. The Government reserves the right to terminate its inspection upon any failure of a fuze, subassembly or component in the sample to comply with any of the specified requirements. In the event of rejection, the Government reserves the right to require the contractor to take corrective action and submit a new first article sample or portion thereof. Until a first article sample is accepted, the contractor is in no way authorized by the Government to initiate regular production (including procurement of material by the contractor) unless otherwise directed by the Contracting Officer.

4.3 Inspection Provisions

4.3.1 Lot Formation. - The term "inspection lot" as used in this specification is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision and complies with the provisions for submission of product as specified in MIL-STD-105. Changes to the process, specification or drawing not affecting safety, performance, interchangeability or storage, as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168. Each lot of fuzes shall contain parts from only one lot interfix number from each supplier and shall use no more than one supplier for each part.

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4.3.2 Examination. - The examinations listed in the following sub-paragraphs shall be performed on inspection lots (see 4.3.1) of the components or assemblies designated in the applicable paragraph headings. Inspection for Critical Defects (and Major Defects, when so specified) shall be 100 percent. Sampling plans for Major and Minor Defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the Government. Instead of applying the AQL's designated herein to the groups of Major and Minor Defects listed, the contractor may elect to apply AQL's to individual defects using an AQL of 0.40 percent for each Major Defect and an AQL of 0.65 percent for each Minor Defect. When this method is selected, it shall be applied to all of the Major and Minor Defects listed in the applicable sub-paragraph of this section, except where 100 percent inspection is specified, and be documented as part of the contractor's inspection system. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

4.3.2.1 Plate, Timer, Number 1 (see dwg. P-9227872).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.65 Percent		
101.	Diameter of holes	Gage	01001
102.	Location of holes	Gage	01002
103.	Flatness	Gage	01003
104.	Surface finish of holes	Gage	01004
Minor:	AQL 0.65 Percent		
201.	Evidence of poor workmanship (see 3.8)	Visual	01005

4.3.2.2 Plate, Timer, Number 2 (see dwg. P-9227873).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.65 Percent		
101.	Location of holes	Gage	02001
102.	Surface finish of holes	Gage	02002
103.	Flatness	Gage	02003
104.	Diameter of holes	Gage	02004

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Minor:	AQL 0.65 Percent		
201.	Evidence of poor workmanship (see 3.8)	Visual	02005

4.3.2.3 Plate, Timer, Number 3 (see dwg. P-9227884).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.65 Percent		
101.	Location of holes	Gage	03001
102.	Surface finish of holes	Gage	03002
103.	Flatness	Gage	03003
104.	Diameter of holes	Gage	03004
Minor:	AQL 0.65 Percent		
201.	Evidence of poor workmanship (see 3.8)	Visual	03005

4.3.2.4 Lock, Rotor (see dwg. P-9227818).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 Percent		
101.	Diameter of head	Gage	04001
Minor:	AQL 0.65 Percent		
201.	Evidence of poor workmanship (see 3.8)	Visual	04002

4.3.2.5 Case, Fuze (see dwg. P-9227866).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		

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Major:	AQL 1.00 Percent		
101.	Spot face thickness	Gage	05001
102.	Diameter of spot face	Gage	05002
103.	Spot face concentric to outside diameter	Gage	05003
104.	Location of hole in base	Gage	05004
105.	Inside diameter, minimum (min.) ...	Gage	05006
106.	Length of inside diameter, min. ...	Gage	05007
Minor:	AQL 0.65 Percent		
201.	Evidence of poor workmanship (see 3.8)	Visual	05005

4.3.2.6 Sleeve, Timer Post (see dwg. P-9227858).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 1.00 Percent		
101.	Outside diameter	Gage	06001
102.	Inside diameter	Gage	06002
103.	Inside diameter concentric to outside diameter	Gage	06003
104.	End of sleeve perpendicular to inside diameter	Gage	06004
105.	End of sleeve parallel to other end	Gage	06005
106.	Length of sleeve	Gage	06006
Minor:	AQL 0.65 Percent		
201.	Evidence of poor workmanship (see 3.8)	Visual	06007

4.3.2.7 Sleeve, Timer Post, Short (see dwg. P-9227859).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 1.00 Percent		
101.	Outside diameter	Gage	07001
102.	Inside diameter	Gage	07002

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103.	Inside diameter concentric to outside diameter	Gage	07003
104.	End of sleeve perpendicular to inside diameter	Gage	07004
105.	End of sleeve parallel to other end	Gage	07005
106.	Length of sleeve	Gage	07006
Minor:	AQL 0.65 Percent		
201.	Evidence of poor workmanship (see 3.8)	Visual	07007

4.3.2.8 Post, Timer (see dwg. P-9227861).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 1.00 Percent		
101.	Outside diameter	Gage	08001
102.	Small outside diameter	Gage	08002
103.	Shaft diameter	Gage	08003
104.	Shaft diameter concentric to outside diameter	Gage	08004
105.	Small diameter concentric to shaft diameter	Gage	08005
106.	Length of shaft	Gage	08006
107.	Width of large diameter	Gage	08007
108.	Width of small diameter	Gage	08008
Minor:	AQL 0.65 Percent		
201.	Evidence of poor workmanship (see 3.8)	Visual	08009

4.3.2.9 Spring, Cam Stop (see dwg. P-9227811).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 1.00 Percent		
101.	Height of bend	Gage	09001
102.	Radius of dimple	Gage	09002

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103.	Location of dimple	Gage	09003
104.	Diameter of hole	Gage	09004
105.	Length of cam spring	Gage	09005
Minor:	AQL 0.65 Percent		
201.	Evidence of poor workmanship (see 3.8)	Visual	09006

4.3.2.10 Pinion, Second Intermediate (see dwg. P-9227798).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 1.00 Percent		
101.	Small outside diameter	Gage	10001
102.	Bottom of large outside diameter perpendicular to small outside diameter	Gage	10002
103.	Burrs on gear teeth	Visual	10003
104.	Length of smallest diameter, both sides	Gage	10004
105.	Distance between smallest diameters	Gage	10005
Minor:	AQL 0.65 Percent		
201.	Evidence of poor workmanship (see 3.8)	Visual	10006

4.3.2.11 Pinion, First Intermediate (see dwg. P-9227799).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 1.00 Percent		
101.	Small outside diameter	Gage	11001
102.	Bottom of large outside diameter perpendicular to small outside diameter	Gage	11002
103.	Burrs on gear teeth	Visual	11003

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104.	Length of smallest diameter, both sides	Gage	11004
105.	Distance between smallest diameters	Gage	11005
Minor:	AQL 0.65 Percent		
201.	Evidence of poor workmanship (see 3.8)	Visual	11006

4.3.2.12 Cam, Rotor (see dwg. P-9227850).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 1.00 Percent		
101.	Diameter of center hole	Gage	12001
102.	Outside diameter	Gage	12002
103.	Diameter of piston actuator hole	Gage	12003
104.	Location of piston actuator hole	Gage	12004
105.	Outside diameter concentric to center hole	Gage	12005
106.	Location of cutaway portion of cam.	Gage	12006
Minor:	AQL 1.0 Percent		
201.	Location of pin hole	Gage	12007
202.	Evidence of poor workmanship (see 3.8)	Visual	12008

4.3.2.13 (Paragraph deleted)

4.3.2.14 Cup, Lead (see dwg. P-9227838).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.65 Percent		
101.	Thickness of bottom of cup	Gage	14001
102.	Outside diameter	Gage	14002
103.	Height	Gage	14003

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Minor:	AQL 0.65 Percent		
201.	Evidence of poor workmanship (see 3.8)	Visual	14004

4.3.2.15 Rotor, Lead (see dwg. P-9227867).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 1.00 Percent		
101.	Outside diameter of shaft	Gage	15001
102.	Outside diameter of shaft below lock ring groove	Gage	15002
103.	Outside diameter of rotor plate mounting boss	Gage	15003
104.	Outside diameter of rotor plate mounting boss concentric to rotor shaft	Gage	15004
105.	Distance from rotor lock ring groove to shaft butting surface ...	Gage	15005
106.	Distance between rotor plate butting surface and top of rotor	Gage	15006
107.	Distance between rotor plate butting surface and base of shaft	Gage	15007
108.	Diameter of lock ring groove	Gage	15008
109.	Location of lead hole	Gage	15009
110.	Thickness of rotor	Gage	15010
Minor:	AQL 1.5 Percent		
201.	Outside diameter of rotor	Gage	15011
202.	Location of tab at end of shaft ...	Gage	15012
203.	Evidence of poor workmanship (see 3.8)	Visual	15013

4.3.2.16 Lock, 5g (see dwg. P-9227856).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		

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Major:	AQL 1.00 Percent		
101.	Outside diameter	Gage	16001
102.	Diameter of head	Gage	16002
103.	Small outside diameter	Gage	16003
104.	Diameter of head concentric to outside diameter	Gage	16004
105.	Diameter of small outside diameter concentric to outside diameter	Gage	16005
106.	Inside diameter	Gage	16006
107.	Height from body to top of head ...	Gage	16007
108.	Height from body to bottom of head	Gage	16008
109.	Length of inside diameter	Gage	16009
110.	Length of large diameter	Gage	16010
Minor:	AQL 0.65 Percent		
201.	Evidence of poor workmanship (see 3.8)	Visual	16011

4.3.2.17 Sleeve, Timer Post, Split (see dwg. P-9227863).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 Percent		
101.	Parallelism of top to bottom	Gage	17001
102.	Size of cutout portion of sleeve ..	Gage	17002
Minor:	AQL 1.5 Percent		
201.	Outside diameter	Gage	17003
202.	Inside diameter	Gage	17004
203.	Evidence of poor workmanship (see 3.8)	Visual	17005

4.3.2.18 Board, Printed Wiring (see dwgs. P-9227894 and P-9227898).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 1.00 Percent		
101.	Outside diameter	Gage	18001
102.	Location of fastening slots	Gage	18002
103.	Diameter of rotor shaft hole	Gage	18003

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104.	Diameter of piston actuator hole ..	Gage		18004
105.	Diameter of detonator hole	Gage		18005
106.	Location of rotor shaft hole	Gage		18006
107.	Location of piston actuator hole ..	Gage		18007
108.	Location of outside diameter	Gage		18008
Minor:	AQL 1.0 Percent			
201.	Thickness	Gage		18009
202.	Evidence of poor workmanship (see 3.8)	Visual		18010

4.3.2.19 Pallet (see dwg. P-9227804).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 Percent		
101.	Location of lever holes	Gage	19001
102.	Diameter of center hole	Gage	19002
Minor:	AQL 0.65 Percent		
201.	Evidence of poor workmanship (see 3.8)	Visual	19003

4.3.2.20 Plate, Lead Rotor (see dwg. P-9227868).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 1.00 Percent		
101.	Location of 5 to 10g rotor lock slot	Gage	20001
102.	Thickness of lip of 5 to 10g rotor lock slot	Gage	20002
103.	Location of tab	Gage	20003
104.	Outside diameter concentric to center hole	Gage	20004
105.	Location of lead hole	Gage	20005
106.	Location of spring hole	Gage	20006
107.	Location of rotor lock hole	Gage	20007

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Minor:	AQL 0.65 Percent		
201.	Evidence of poor workmanship (see 3.8)	Visual	20008

4.3.2.21 Cover (see dwg. P-9227791).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 Percent		
101.	Outside diameter, min.	Gage	21001
102.	Location of hole	Gage	21002
Minor:	AQL 1.5 Percent		
201.	Thickness	Gage	21003
202.	Diameter of hole	Gage	21004
203.	Evidence of poor workmanship (see 3.8)	Visual	21005

4.3.2.22 Shaft, Pallet (see dwg. P-9227803).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 1.00 Percent		
101.	Bottom of large outside diameter perpendicular to small diameter	Gage	22001
102.	Tapered shaft concentric to small outside diameter	Gage	22002
103.	Smallest diameter, both sides	Gage	22003
104.	Length of smallest diameter, both sides	Gage	22004
105.	Distance between smallest diameters	Gage	22005
Minor:	AQL 0.65 Percent		
201.	Evidence of poor workmanship (see 3.8)	Visual	22006

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4.3.2.23 Shaft, Drive Gear (see dwg. P-9227807).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 1.00 Percent		
101.	Datum A diameter	Gage	23001
102.	Large diameter	Gage	23002
103.	Datum B diameter	Gage	23003
104.	Datum C diameter	Gage	23004
105.	Perpendicularity of shoulder	Gage	23005
106.	Datum B concentric to Datum A	Gage	23006
107.	Surface finish of Datum A and B	Gage	23007
108.	Datum C concentric to Datum A and B	Gage	23008
Minor:	AQL 0.65 Percent		
201.	Evidence of poor workmanship (see 3.8)	Visual	23009

4.3.2.24 Body (see dwg. P-9227904).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 1.00 Percent		
101.	Location of detonator cavity	Gage	24001
102.	Thickness of lip at end of detonator cavity	Gage	24002
103.	Diameter of boost switch cavity	Gage	24003
104.	Location of timer post holes	Gage	24004
105.	Diameter of 5g lock cavity	Gage	24005
106.	Location of piston actuator hole	Gage	24006
107.	Location of printed circuit board holes	Gage	24007
Minor:	AQL 0.65 Percent		
201.	Evidence of poor workmanship (see 3.8)	Visual	24008

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4.3.2.25 Pallet Assembly (see dwg. P-9227805).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.65 Percent		
101.	Lubricant missing	Visual	25001
102.	Distance of pallet lever from bottom of pallet	Gage	25002
103.	Length from bottom of pallet to end of shaft, min.	Gage	25003
Minor:	AQL 0.65 Percent		
201.	Evidence of poor workmanship (see 3.8)	Visual	25004

4.3.2.26 Escapement Wheel and Pinion Assembly (see dwg. P-9227813).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 Percent		
101.	Length from bottom of escapement wheel to end of shaft, maximum (max.)	Gage	26001
102.	Lubricant missing	Visual	26002
Minor:	AQL 0.65 Percent		
201.	Evidence of poor workmanship (see 3.8)	Visual	26003

4.3.2.27 Boost Switch Plug Assembly (see dwg. P-9227795).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 Percent		
101.	Crimp not 360 degrees	Visual	27001

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Minor:	AQL 0.65 Percent		
201.	Evidence of poor workmanship (see 3.8)	Visual	27002

4.3.2.28 Printed Wiring Board Assembly (see dwgs. P-9227893 and P-9227900).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 1.00 Percent		
101.	Height of components, max.	Gage	28001
102.	Height of solder on connections, max.	Gage	28002
103.	Location of header	Gage	28003
104.	Varnish outside of specified area	Visual	28004
105.	Components not soldered properly	Visual	28005
106.	Insulating cover on MMS switch not 90 to 180° from printed wiring board	Visual	28006
107.	Rotary switch assembled improperly.	Visual	28007
Minor:	AQL 0.65 Percent		
201.	Evidence of poor workmanship (see 3.8)	Visual	28008

4.3.2.29 Body Assembly (see dwgs. P-9227890 and P-9227903).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 Percent		
101.	Plugs improperly welded	Visual	29001
102.	Screws not secured by adhesive	Visual	29002
Minor:	AQL 0.65 Percent		
201.	Evidence of poor workmanship (see 3.8)	Visual	29003

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4.3.2.30 Wire Assembly (see dwgs. P-9227843, P-9227844, P-9227845 and P-9227846).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 Percent		
101.	Microsocket not staked properly ...	Visual	30001
Minor:	AQL 0.65 Percent		
201.	Evidence of poor workmanship (see 3.8)	Visual	30002

4.3.2.31 Rotor Lead Assembly (see dwg. P-9227870).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.65 Percent		
101.	Location of rotor plate	Gage	31001
102.	Distance between rotor and plate, max.	Gage	31002
103.	Plate not staked properly	Gage	31003
104.	Lead cup assembly improperly assembled	Visual	31004
Minor:	AQL 0.65 Percent		
201.	Evidence of poor workmanship (see 3.8)	Visual	31005

4.3.2.32 Lead Cup Assembly (see dwg. P-9227839).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.65 Percent		
101.	Outside diameter	Gage	32001
102.	Height, max.	Gage	32002
103.	Lead consolidated improperly	Visual	32003

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Minor:	AQL 0.65 Percent		
201.	Evidence of poor workmanship (see 3.8)	Visual	32004

4.3.2.33 Timer Assembly (see dwg. P-9227891).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.65 Percent		
101.	Parts assembled improperly	Visual	33001
102.	Cam pin not engaging stop spring	Visual	33002
103.	Distance between bottom of cam and bottom of cam pin	Gage	33003
Minor:	AQL 0.65 Percent		
201.	Evidence of poor workmanship (see 3.8)	Visual	33004

4.3.2.34 Mechanism Assembly (see dwg. P-9227889).

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Mechanism assembly not locked in unarmed position	Visual	34001
Major:	AQL 1.00 Percent		
101.	Cement missing from components	Visual	34002
102.	Cam switch screw not sealed with sealing compound	Visual	34003
103.	Screws holding printed wiring board to body not melted	Visual	34004
104.	Connections not soldered properly	Visual	34005
105.	Distance from top of printed wiring board to top of body, max.	Gage	34006
106.	Epoxy not on switch terminal as specified	Visual	34007
107.	Switch wires not connected properly	Visual	34008

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Minor:	AQL 0.65 Percent		
201.	Evidence of poor workmanship (see 3.8)	Visual	34009

4.3.2.35 M814 Fuze Assembly, Prior to Installation of Cover (see dwg. P-9227896).

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Mechanism assembly not locked in unarmed position	Visual	35001
Major:	None defined.		
Minor:	AQL 0.65 Percent		
201.	Evidence of poor workmanship (see 3.8)	Visual	35002

4.3.2.36 M814 Fuze Assembly (see dwg. P-9227896).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 1.00 Percent		
101.	Outside diameter, max.	Gage	36001
102.	Length, max.	Gage	36002
103.	Crimp not 360 degrees	Visual	36003
104.	Location of connectors	Gage	36004
105.	Length of wires, min.	Gage	36005
106.	Height of connectors and wires, max	Gage	36006
107.	Connectors, wires or epoxy outside of envelope area	Gage	36007
Minor:	AQL 1.0 Percent		
201.	Evidence of poor workmanship (see 3.8)	Visual	36008
202.	Marking missing, misleading or unidentifiable	Visual	36009

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4.3.2.37 Rotor Assembly (TM only) (see dwg. P-9227880).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.65 Percent		
101.	Location of rotor plate	Gage	37001
102.	Distance between rotor and plate, max.	Gage	37002
103.	Plate not staked properly	Gage	37003
Minor:	AQL 0.65 Percent		
201.	Evidence of poor workmanship (see 3.8)	Visual	37004

4.3.2.38 Mechanism Assembly (TM only) (see dwg. P-9227901).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 1.00 Percent		
101.	Mechanism assembly not locked in unarmed position	Visual	38001
102.	Cam switch screw not sealed with sealing compound	Visual	38002
103.	Screws holding printed wiring board to body not melted	Visual	38003
104.	Connections not soldered properly .	Visual	38004
105.	Cement missing from components	Visual	38005
106.	Distance from top of printed wiring board to top of body, max. .	Gage	38006
107.	Epoxy not on switch terminal as specified	Visual	38007
108.	Switch wires not connected properly	Visual	38008
Minor:	AQL 0.65 Percent		
201.	Evidence of poor workmanship (see 3.8)	Visual	38009

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4.3.2.39 M814 Fuze Assembly (TM only) (see dwg. P-9227902).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 1.00 Percent		
101.	Outside diameter, max.	Gage	39001
102.	Length, max.	Gage	39002
103.	Crimp not 360°	Visual	39003
104.	Location of connector	Gage	39004
105.	Length of wires, min.	Gage	39005
106.	Height of connector and wires, max.	Gage	39006
107.	Connector, wires or epoxy outside of envelope area	Gage	39007
Minor:	AQL 1.0 Percent		
201.	Evidence of poor workmanship (see 3.8)	Visual	39008
202.	Marking missing, misleading or unidentifiable	Visual	39009

4.3.2.40 Contact, Rotary Switch (see dwg. P-9227792).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.65 Percent		
101.	Length	Gage	40001
102.	Width	Gage	40002
103.	Overall height	Gage	40003
Minor:	AQL 0.65 Percent		
201.	Evidence of poor workmanship (see 3.8)	Visual	40004

4.3.2.41 Housing, Rotary Switch (see dwg. P-9227793).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:16Z
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Major:	AQL 0.40 Percent		
101.	Location of legs	Gage	41001
102.	Inside diameter	Gage	41002
Minor:	AQL 0.65 Percent		
201.	Evidence of poor workmanship (see 3.8)	Visual	41003

4.3.2.42 Pinion, Escapement (see dwg. P-9227806).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.65 Percent		
101.	Smallest diameter, both sides	Gage	42001
102.	Length of smallest diameter, both sides	Gage	42002
103.	Distance between smallest diameter	Gage	42003
Minor:	AQL 0.65 Percent		
201.	Evidence of poor workmanship (see 3.8)	Visual	42004

4.3.2.43 Wheel, Escapement (see dwg. P-9227812).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 Percent		
101.	Center hole concentric to diameter	Gage	43001
102.	Configuration of teeth	Gage	43002
Minor:	AQL 0.65 Percent		
201.	Evidence of poor workmanship (see 3.8)	Visual	43003

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4.3.2.44 Rotor, Rotary Switch (see dwg. P-9227869).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 1.00 Percent		
101.	Location of slots	Gage	44001
102.	Outside diameter concentric to inside diameter	Gage	44002
103.	Length of slot	Gage	44003
104.	Width of slot	Gage	44004
105.	Outside diameter	Gage	44005
Minor:	AQL 0.65 Percent		
201.	Evidence of poor workmanship (see 3.8)	Visual	44006

4.3.3 Testing

4.3.3.1 First Article Sample Testing (see 4.2.1). - The samples submitted as the first article quantity shall be tested as specified herein after being examined 100 percent for the applicable classifications of defects in 4.3.2. Inspection equipment used shall be as specified in 4.3.4. Rejection of the first article sample for failure of any component or assembly in any of the testing specified herein shall be as specified in 4.2.3.

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4.3.3.1.1 Mechanism Assembly (P-9227889), Less Detonator and Piston Actuator.

Code No.	Category	Sample Size	Defect	Requirement	Test Method
34010	Critical	5	Assembly arms during Non-Arming Test	3.3.3	4.4.1
34011	Major	5	Assembly fails to arm during Arming Test	3.3.3	4.4.1
34012	Major	5	Assembly fails Sustainer Time Test	3.3.2	4.4.2
34013	Major	5	Assembly fails Arming Time Test	3.3.3	4.4.2
34014	Major	5	Boost Switch closes during Non-Functioning Test	3.3.1.2	4.4.3
34015	Major	5	Boost Switch fails to close during Functioning Test	3.3.1.2	4.4.3
34016	Major	5	Impact Switch fails to function during Functioning Test	3.3.7	4.4.4
34017	Major	5	Tubular Rivets (P-9227814) fail Push Test	P-9227889	4.4.18

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4.3.3.1.2 Mechanism Assembly (P-9227889).

Code No.	Category	Sample Size	Defect	Requirement	Test Method
34018	Major	5	Assembly fails Circuit Continuity Test	3.4	4.4.15
34019	Major	5	Gap between Timing Cam and Rotor Tab	P-9227889	Gage
34020	Major	5	Distance from top of Cam to Piston Actuator Plunger	P-9227889	Gage
34021	Major	5	Epoxy not on Microswitch Terminal	P-9227889	Visual
34022	Major	5	Epoxy not locking Timing Cam	P-9227889	Visual
34023	Major	5	Solder connections improper	P-9227889	Visual
34024	Major	5	Surface finish improper or burrs on functional side of Tab	P-9227868	Visual

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4.3.3.1.3 Printed Wiring Board Assembly (P-9227893).

Code No.	Category	Sample Size	Defect	Requirement	Test Method
28009	Major	5	Assembly fails Self-Destruct Circuit Test	3.3.6	4.4.5
28010	Major	5	Rotor Switch fails Continuity Test	4.4.14	4.4.14

4.3.3.1.4 Pallet Assembly (P-9227805).

Code No.	Category	Sample Size	Defect	Requirement	Test Method
25005	Major	5	Assembly fails Push Test	P-9227805	4.4.7

4.3.3.1.5 Electric Detonator, M95 (9205891).

Code No.	Category	Sample Size	Defect	Requirement	Test Method
45001	Major	5	Detonator fails Resistance Test	3.3.9	4.4.8

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4.3.3.1.6 Piston Actuator (P-9227817).

Code No.	Category	Sample Size	Defect	Requirement	Test Method
46001	Major	5	Actuator fails Resistance Test	3.3.7	4.4.9
46002	Major	5	Actuator fails Explosive Functioning Test	P-9227817	4.4.16

4.3.3.1.7 MMS 1.0 Switch (P-9227874).

Code No.	Category	Sample Size	Defect	Requirement	Test Method
47001	Major	5	Switch fails Resistance Test	3.3.1.1	4.4.10
47002	Major	5	Switch fails Explosive Functioning Test	P-9227874	4.4.17

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4.3.3.1.8 Molded Components Tests

Code No.	Category	Sample Size	Defect	Requirement	Test Method
48001	All: Major	Each: 5 pieces per cavity per mold	Transistor Mount fails any drawing requirement	P-9227837	All: Gages as needed to check drawing requirements
48002			Rotary Switch Rotor fails any drawing requirement	P-9227869	
48003			Rotary Switch Housing fails any drawing requirement	P-9227793	
48004			Feed Thru Header fails any drawing requirement	P-9227808	
48005			Detonator Plug fails any drawing requirement	P-9227827	
48006			G-Lock Plug fails any drawing requirement	P-9227822	
48007			Impact Switch Plug fails any drawing requirement	P-9227832	
48008			Boost Switch Plug fails any drawing requirement	P-9227826	
48009			Switch Cam fails any drawing requirement	P-9227819	

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4.3.3.1.9 Spring Embrittlement Tests

Code No.	Category	Sample Size	Defect	Requirement	Test Method
49001	Major	5	Timer Power Spring (P-9227871) breaks	3.9	4.4.11
49002	Major	5	5-G Lock Spring (P-9227853) breaks	3.9	4.4.11
49003	Major	5	Rotary Switch Spring (P-9227810) breaks	3.9	4.4.11
49004	Major	5	Boost Switch Spring (P-9227802) breaks	3.9	4.4.11
49005	Major	5	Impact Switch Spring (P-9227815) breaks	3.9	4.4.11
49006	Major	5	Rotor Spring (P-9227851) breaks	3.9	4.4.11
49007	Major	5	Rotor Lock Spring (P-9227796) breaks	3.9	4.4.11

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4.3.3.1.10 Spring Load Tests

Code No.	Category	Sample Size	Defect	Requirement	Test Method
50001	Major	5	Rotor Lock Spring fails at either position	P-9227796	4.4.12
50002	Major	5	Impact Switch Spring fails at either position	P-9227815	4.4.12
50003	Major	5	Boost Switch Spring fails at either position	P-9227802	4.4.12
50004	Major	5	Rotary Switch Spring fails at either position	P-9227810	4.4.12
50005	Major	5	5-G Lock Spring fails at either position	P-9227853	4.4.12

4.3.3.1.11 Spring Torque Tests

Code No.	Category	Sample Size	Defect	Requirement	Test Method
51001	Major	5	Timer Power Spring fails at either position	P-9227871	4.4.13
51002	Major	5	Rotor Spring fails at either position	P-9227851	4.4.13

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4.3.3.1.12 Fuze Assembly, M814 (P-9227896)

Code No.	Category	Sample Size	Defect	Requirement	Test Method
36010	Major	12	Fuze fails Circuit Continuity Test	3.4	4.4.15
36011	Major	Table I	Fuze fails Environmental Testing	Table I and 3.5	4.4.19

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4.3.3.1.13 Mechanism Assembly (TM only) - (P-9227901).

Code No.	Category	Sample Size	Defect	Requirement	Test Method
38010	Major	5	Telemetry Signal Test	4.4.6	4.4.6
38011	Major	5	Gap between Timing Cam and Rotor Tab	P-9227901	Gage
38012	Major	5	Distance from top of Cam to Piston Actuator Plunger	P-9227901	Gage
38013	Major	5	Epoxy not on Micro-Switch Terminal	P-9227901	Visual
38014	Major	5	Epoxy not locking Timing Cam	P-9227901	Visual
38015	Major	5	Solder connection improper	P-9227901	Visual
38016	Major	5	Surface finish or burrs on functional side of Tab	P-9227868	Visual

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4.3.3.1.14 M814 Fuze Assembly (TM only) - (P-9227902).

Code No.	Category	Sample Size	Defect	Requirement	Test Method
39010	Major	12	Fuze fails Environmental Testing	Table I and 3.5	4.4.20

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TABLE I

ENVIRONMENTAL TEST SCHEDULE

Sample Size	Temperature °F ± 5°F	Number of Samples Functioned					
		Group I		Group II		Group III	
		IMP	SD	IMP	SD	IMP	SD
12	- 45	1	1	1	1	1	1
	+140	1	1	1	1	1	1

Code:

Environments (requirement paragraphs): Group I = (3.5.5) + (3.5.1)
Group II = (3.5.2) + (3.5.1)
Group III = (3.5.2) + (3.5.5)

Functioning Requirements:

IMP = Switch, Impact (3.3.5)
SD = Self Destruct (3.3.6)

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4.3.3.1.15 Jolt and Jumble Test (see 3.6)

4.3.3.1.15.1 Before Functioning Piston Actuator - Fuze fails test - Critical Defect - Code No. 36012. - This test shall be performed on a sample of two (2) fuzes (dwg. P-9227896) which have been assembled to inert warheads or simulated warheads. The assemblies shall be subjected to the jolt test specified in MIL-STD-331, Test 101 and then, without disassembly of the fuze, to the jumble test specified in MIL-STD-331, Test 102. The fuzes shall be disassembled for inspection only after completion of the jumble test. The first article sample shall be rejected if any fuze has functioned, if a piston actuator has failed, if a fuze is armed or if any fuze is unsafe to handle.

4.3.3.1.15.2 After Functioning Piston Actuator - Fuze fails test - Critical Defect - Code No. 36013. - A second sample of two (2) fuzes (dwg. P-9227896) shall be subjected to the successive jolt and jumble testing described in 4.3.3.1.15.1 except that each piston actuator (dwg. P-9227817) shall be functioned prior to performance of the jolt and jumble tests so that the only restraint on the rotor during the test will be the 5g lock (dwg. P-9227856). The fuzes previously tested in 4.3.3.1.15.1 shall not be used for this test. The first article sample shall be rejected if any fuze has functioned, if a 5g lock has failed, if a fuze is armed or if any fuze is unsafe to handle.

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4.3.3.2 Production Testing

4.3.3.2.1 Testing of Mechanism Assembly (dwg. 9227889), prior to assembly of detonator (dwg. 9205891) and piston actuator (dwg. 9227817). - The following tests shall be performed 100 percent (except where noted) using inspection equipment in accordance with 4.3.4. Any assembly which fails to satisfy the requirements shall be removed from the lot.

Test Para. No.	Test Description and Defect	Category	Defect Code No.	Requirement Para. No.	Test Method Para. No.
4.3.3.2.1.1	Non-Arming Test at 5g's			3.3.3	4.4.1
	Assembly arms during test	Critical	34010		
4.3.3.2.1.2	Arming Test at 10g's			3.3.3	4.4.1
	Assembly fails to arm	Major	34011		
4.3.3.2.1.3	Arming & Sustainer Time Test				4.4.2
	Assembly fails Sustainer Time Test	Major	34012	3.3.2	
	Assembly fails Arming Time Test	Major	34013	3.3.3	
4.3.3.2.1.4	Boost Switch Test			3.3.1.2	4.4.3
	Switch closes during Non-Functioning Test	Major	34014		
	Switch fails to close during Functioning Test	Major	34015		
4.3.3.2.1.5	Impact Switch Test*			3.3.5	4.4.4
	Switch fails to function during Functioning Test	Major	34016		
4.3.3.2.1.6	Push Test on Sustainer Switch				
	Any rivet deforms or moves	Major	34017	P-9227889	4.4.18

* Perform test on a sample of 32 assemblies first. Acceptance number shall be zero (0) defectives. If a defective is found, test 100 percent.

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4.3.3.2.2 Testing of Complete Mechanism Assembly (dwg. 9227889). - he following tests shall be performed 100 percent on mechanism assemblies using inspection equipment in accordance with 4.3.4. Any assembly which fails to satisfy the specified requirements shall be removed from the lot.

Test Para. No.	Test Description and Defect	Category	Defect Code No.	Requirement Dwg. No.	Test Method
4.3.3.2.2.1	Gap Between Timing Cam & Rotor Tab	Major	34019	9227889	Gage
4.3.3.2.2.2	Distance from top of Timing Cam to Piston Actuator Plunger	Major	34020	9227889	Gage
4.3.3.2.2.3	Epoxy not on MMS Terminal	Major	34021	9227889	Visual
4.3.3.2.2.4	Epoxy not locking Timing Cam	Major	34022	9227889	Visual
4.3.3.2.2.5	Solder Connection improper	Major	34023	9227889	Visual
4.3.3.2.2.6	Surface finish improper or burrs on functional side of Tab	Major	34024	9227868	Visual

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4.3.3.2.3 Printed Wiring Board Assembly (P-9227893).

Code No.	Category	Sample Size	Defect	Requirement	Test Method
28009	Major	100%	Assembly fails Self-Destruct Circuit Test	3.3.6	4.4.5
28010	Major	100%	Rotor Switch fails Continuity Test	4.4.14	4.4.14

4.3.3.2.4 Pallet Assembly (P-9227805).

Code No.	Category	Sample Size	Defect	Requirement	Test Method
25005	Major	a.	Assembly fails Push Test	P-9227805	4.4.7

4.3.3.2.5 Electric Detonator, M95 (9205891).

Code No.	Category	Sample Size	Defect	Requirement	Test Method
45001	Major	100%	Detonator fails Resistance Test	3.3.9	4.4.8

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a. MIL-STD-105, Inspection Level II, Table II-A, AQL 0.65. After three consecutive successful lots, change to Level I.

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4.3.3.2.6 Piston Actuator (P-9227817).

Code No.	Category	Sample Size	Defect	Requirement	Test Method
46001	Major	100%	Actuator fails Resistance Test	3.3.7	4.4.9
46002	Major	b.	Actuator fails Explosive Functioning Test	P-9227817	4.4.16

4.3.3.2.7 M.M.S. 1.0 (P-9227874).

Code No.	Category	Sample Size	Defect	Requirement	Test Method
47001	Major	100%	Switch fails Resistance Test	3.3.1.1	4.4.10
47002	Major	b.	Switch fails Explosive Functioning	P-9227874	4.4.17

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b. For lot size 0 - 500, sample = 50, acceptance number = 0.

For lot size greater than 500, sample = 10 percent of lot, acceptance number = 0.

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4.3.3.2.8 Molded Components Tests

Code No.	Category	Sample Size	Defect	Requirement	Test Method
48001	All: Major	a.	Transistor Mount fails any drawing requirement	P-9227837	All: Gages as needed to check drawing requirements
48002		a.	Rotary Switch Rotor fails any drawing requirement	P-9227869	
48003		a.	Rotary Switch Housing fails any drawing requirement	P-9227793	
48004		a.	Feed Thru Header fails any drawing requirement	P-9227808	
48005		a.	Detonator Plug fails any drawing requirement	P-9227827	
48006		a.	G-Lock Plug fails any drawing requirement	P-9227822	
48007		a.	Impact Switch Plug fails any drawing requirement	P-9227832	
48008		a.	Boost Switch Plug fails any drawing requirement	P-9227826	
48009		a.	Switch Cam fails any drawing requirement	P-9227819	

a. Inspect the first five pieces per cavity per mold for all drawing requirements. If a failure occurs, correct the defective mold and resample from the corrected mold. When five pieces from each cavity per mold have met all requirements, commence production. After 1000 pieces from each cavity per mold have been produced, reverify the molds.

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4.3.3.2.9 Spring Embrittlement Tests

Code No.	Category	Sample Size	Defect	Requirement	Test Method
49001	Major	a.	Timer Power Spring (P-9227871) breaks	3.9	4.4.11
49002	Major	a.	5-G Lock Spring (P-9227853) breaks	3.9	4.4.11
49003	Major	a.	Rotary Switch Spring (P-9227810) breaks	3.9	4.4.11
49004	Major	a.	Boost Switch Spring (P-9227802) breaks	3.9	4.4.11
49005	Major	a.	Impact Switch Spring (P-9227815) breaks	3.9	4.4.11
49006	Major	a.	Rotor Spring (P-9227851) breaks	3.9	4.4.11
49007	Major	a.	Rotor Lock Spring (P-9227796) breaks	3.9	4.4.11

a. Sample size shall be ten (10) springs. Acceptance number for each test sample of 10 springs shall be zero (0) defectives.

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4.3.3.2.10 Spring Load Tests

Code No.	Category	Sample Size	Defect	Requirement	Test Method
50001	Major	a.	Rotor Lock Spring fails at either position	P-9227796	4.4.12
50002	Major	a.	Impact Switch Spring fails at either position	P-9227815	4.4.12
50003	Major	a.	Boost Switch Spring fails at either position	P-9227802	4.4.12
50004	Major	a.	Rotary Switch Spring fails at either position	P-9227810	4.4.12
50005	Major	a.	5-G Lock Spring fails at either position	P-9227853	4.4.12

4.3.3.2.11 Spring Torque Tests

Code No.	Category	Sample Size	Defect	Requirement	Test Method
51001	Major	a.	Timer Power Spring fails at either position	P-9227871	4.4.13
51002	Major	a.	Rotor Spring fails at either position	P-9227851	4.4.13

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a. Sample size shall be ten (10) springs. Acceptance number for each test sample of 10 springs shall be zero (0) defectives.

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4.3.3.2.12 Fuze Assembly, M814 (P-9227896).

Code No.	Category	Sample Size	Defect	Requirement	Test Method
36010	Major	100%	Fuze fails Circuit Continuity Test	3.4	4.4.15
36011	Major	Table II a.	Fuze fails Environmental Testing	Table II and 3.5	4.4.19

a. Divide the approximate lot size by three (3) to form three production groups. These groups are identified as I, II and III in Table II. As production of Group I is completed, randomly select the required samples and test per Table II. Follow the same procedure for Groups II and III. All assemblies tested must pass all timing requirements and function high order for the lot to be accepted.

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4.3.3.2.13 Mechanism Assembly (TM Only) - (P-9227901).

Code No.	Category	Sample Size	Defect	Requirement	Test Method
38010	Major	100%	Telemetry Signal Test	4.4.6	4.4.6
38011	Major	100%	Gap between Timing Cam and Rotor Tab	P-9227901	Gage
38012	Major	100%	Distance from top of Cam to Piston Actuator Plunger	P-9227901	Gage
38013	Major	100%	Epoxy not on Micro-Switch Terminal	P-9227901	Visual
38014	Major	100%	Epoxy not locking Timing Cam	P-9227901	Visual
38015	Major	100%	Solder connection improper	P-9227901	Visual
38016	Major	100%	Surface finish or burrs on functional side of Tab	P-9227868	Visual

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4.3.3.2.14 M814 Fuze Assembly (TM Only) - (P-9227902).

Code No.	Category	Sample Size	Defect	Requirement	Test Method
39010	Major	Table II a.	Fuze fails Environmental Testing	Table II and 3.5	4.4.20

a. Divide the approximate lot size by three (3) to form three production groups. These groups are identified as I, II and III in Table II. As production of Group I is completed, randomly select the required samples and test per Table II. Follow the same procedure for Groups II and III. All assemblies tested must pass all timing and functioning requirements for the lot to be accepted.

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TABLE II

ENVIRONMENTAL TEST SCHEDULE

Lot Size	Sample Size	Temperature °F ± 5°F	Number of Samples Functioned					
			Group I		Group II		Group III	
			IMP	SD	IMP	SD	IMP	SD
1 - 50	9	- 45	1	1	-	1	1	1
		+140	-	1	1	1	-	1
51 - 125	12	- 45	1	1	1	1	1	1
		+140	1	1	1	1	1	1
126 - 275	18	- 45	1	2	1	2	1	2
		+140	1	2	1	2	1	2
276 - 600	24	- 45	2	2	2	2	2	2
		+140	2	2	2	2	2	2

Code:

Environments (Requirement Paragraphs): - Group I = (3.5.5) + (3.5.1)
Group II = (3.5.2) + (3.5.1)
Group III = (3.5.2) + (3.5.5)

Functioning Requirements:

IMP = Switch, Impact (3.3.5)
SD = Self Destruct (3.3.6)

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4.3.4 Inspection Equipment (see 6.2 and 6.3). - The inspection equipment required to perform the examinations and tests prescribed in 4.3.2 is identified under the "Method of Inspection" headings. Unless otherwise specified, the inspection equipment used to perform the examinations and tests prescribed in 4.3.3 shall be contractor designed (see 6.3.2)

4.4 Test Methods and Procedures

4.4.1 Non-Arming and Arming Test. - The assembly shall be positioned in the centrifuge test fixture, simulating the position of the fuze in its case, and shall be held in the unarmed position by a pin which simulates the retractable piston actuator. When a constant force of 5g's (longitudinal acceleration) is established, the pin shall be released and the assembly shall not arm (see 6.5). The assembly shall then be held in the unarmed position by the restraining pin; and when a constant force of 10g's is applied, the pin shall be released and the assembly shall arm (see 6.6). Assemblies which arm during the non-arming test or do not arm during the arming test shall be classed defective.

4.4.2 Arming and Sustainer Time Test. - This test is performed using mechanism assemblies (dwg. 9227889) prior to installation of the detonator (dwg. 9205891) and piston actuator (dwg. 9227817). The assembly shall be positioned in the centrifuge test fixture simulating the position of the fuze in its case and held in the unarmed position by a pin which simulates the retractable piston actuator. The sustainer time and arm time shall be set at a time to be established by the contractor in order to meet the requirements of 3.3.2 and 3.3.3. The set times shall be within a tolerance of $\pm$ .003 seconds for sustainer time and $\pm$ .15 seconds for arming time when measured at 33g's on the centrifuge. During production, two (2) additional runs at 33g's shall be made before final cam lock and one (1) run at 33g's after cam lock and melting the screws on the printed wiring board. For these three runs, the assembly shall be classed defective if the sustainer time is not within $\pm$ .017 seconds of the set value, or if the arming time is not within $\pm$ .15 seconds of its set value. All values of time shall be recorded. When the test is applied to a first article sample, the final cam lock and melting of screws will have preceded the sample submission and only the final test run will be performed during first article testing. However, all contractor timing data associated with the sample shall be submitted with the sample for use by the Government in evaluation of the sample.

4.4.3 Boost Switch Test. - The assembly shall be positioned in the centrifuge test fixture which simulates the position of the fuze in the case. The centrifuge shall be run at 23g's minimum longitudinal acceleration, and the action of the boost switch shall be monitored on an oscilloscope. If the boost switch closes, the assembly shall be classed defective. The speed of the centrifuge shall be increased to 33g's maximum longitudinal acceleration; and if the boost switch remains open, the assembly shall be classed defective.

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4.4.4 Impact Switch Test. - The impact switch shall be functioned in a suitable test fixture by application of an impact shock equivalent to 33g's maximum for a minimum duration of six (6) milliseconds. Functioning shall be defined as a closed-circuit reading across the impact switch contacts. Any switch which fails to function shall be classed defective.

4.4.5 Self-Destruct Circuit Test. - The self-destruct circuit shall be tested twice: once with a 1K resistor simulating the detonator and once with a 3.3K resistor simulating the detonator. The printed wiring board assembly shall have a minimum peak voltage of 12 volts and shall be classed defective if the functioning time does not satisfy the requirement or if the peak voltage is less than the minimum required.

4.4.6 Telemetry Signal Test. - The telemetry (TM) mechanism assembly shall be armed and shall exhibit an R-C network signal with a trailing edge decay upon arming that has a peak voltage of not less than 40 volts and a time constant of 30 milliseconds. The TM mechanism assembly shall exhibit a signal similar in shape to the arming pulse, but opposite in polarity at a time that will assure the assembly's meeting the system requirements for the self-destruct signal. The assembly shall be classed defective if it fails to satisfy any of the above requirements.

4.4.7 Push Test on Pallet Assembly. - A force as specified in the applicable drawing shall be applied to the lever pallets in the reverse direction of assembly. The pallet assembly shall be classed defective if any pallet fails to withstand the applied force.

4.4.8 Detonator Resistance Test. - The resistance of the M95 electric detonator shall be measured using a meter with current limited to 20 microamperes maximum. The test should be conducted in an explosive protected area with explosive handling safety procedures in effect. The detonator shall be classed defective if the resistance is less than 1,000 ohms or more than 10,000 ohms.

4.4.9 Piston Actuator Resistance Test. - The resistance of the piston actuator shall be measured using a meter with current limited to 10 milliamperes maximum. The test should be conducted in an explosive protected area with explosive handling safety procedures in effect. The piston actuator shall be classed defective if its resistance is less than 1.4 ohms or more than 2.6 ohms.

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4.4.10 MMS 1.0 Switch Resistance Test. - The resistance of the interlock switch (MMS 1.0) shall be measured using a meter with current limited to five (5) milliamperes maximum. The test should be conducted in an explosive protected area with explosive handling safety procedures in effect. The MMS 1.0 switch shall be classed defective if its resistance is less than five (5) ohms or more than nine (9) ohms.

4.4.11 Spring Embrittlement Test. - The spring shall be extended along its longitudinal axis to the minimum point of permanent distortion. The spring shall be classed defective if it breaks using the minimum distortion load.

4.4.12 Spring Load Test. - The spring shall be tested twice by compressing the spring successively to each of the compressed lengths specified on the drawing and the applicable load for each length shall be measured. The spring shall be classed defective if either applied load is not within the requirements of the applicable drawing.

4.4.13 Spring Torque Test. - The torque shall be measured at each of the specified spring positions when the spring is unwound as specified on the applicable drawing. The spring shall be classed defective if any value of torque at any specified position fails to satisfy the drawing requirement.

4.4.14 Printed Wiring Board Assembly Continuity Test. - The continuity of the rotor switch in the printed wiring board assembly shall be measured with the switch in the armed and the unarmed position using an appropriate ohmeter. The assembly shall be classed defective if the resistance across the rotor switch shows more than one (1) ohm in the armed position or less than one (1) megohm in the unarmed position.

4.4.15 Circuit Continuity Test. - The assembly shall be connected to a continuity tester which simulates the mating connection present in the missile. The test should be conducted in an explosive loading area with explosive handling safety procedures in effect. The electrical continuity shall be measured between all the pairs of points specified in the requirement paragraph. The assembly shall be classed defective if the measurement specified for any pair of points is exceeded.

4.4.16 Piston Actuator Explosive Functioning Test. - The sample of piston actuators shall be divided in half for testing at two temperatures. If the original sample size is an odd number, the odd actuator shall be tested at the lower conditioning temperature. Half the actuators shall be conditioned at -45°F $\pm$ 5°F for a minimum of two (2) hours and the remaining half shall be conditioned at +140°F $\pm$ 5°F for a minimum of two (2) hours. Within three (3) minutes from the time of removal from the conditioning chamber, the piston actuators shall be connected to the firing circuit and

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functioned. The time to function after application of the required current and the length of piston travel shall be recorded on a test data sheet. The piston actuator shall be classed defective if the functioning time is greater than 10 milliseconds or if the minimum travel of the piston against a one (1) pound load minimum is less than the drawing requirement.

4.4.17 MMS 1.0 Explosive Functioning Test. - The sample of micro-miniature switches shall be divided in half for testing at two temperatures. If the original sample size is an odd number, the odd switch shall be tested at the lower conditioning temperature. Half the switches shall be conditioned at -45°F ± 5°F for a minimum of two (2) hours and the remaining half shall be conditioned at +140°F ± 5°F for a minimum of two (2) hours. Within three (3) minutes from the time of removal from the conditioning chamber, the switches shall be connected to the firing circuit and functioned. The firing circuit shall deliver a one (1) ampere pulse of one millisecond duration when in series with a five (5) ohm resistance. The functioning time shall be measured and recorded on a test data sheet. The micro-miniature switch shall be classed defective if it fails to function or if the functioning time is greater than four (4) milliseconds.

4.4.18 Push Test on Sustainer Switch. - A minimum unseating force of ten (10) pounds shall be applied, in the axial direction, to each rivet in the sustainer switch assembly in the mechanism assembly. The assembly shall be classed defective if any rivet moves or is deformed by the applied force.

4.4.19 Environmental Testing (Tactical Fuze). - The specified number of samples shall be assembled to simulated warheads and then subjected to the required environmental conditions in the order shown for the specified environmental group (I, II or III). After completion of environmental conditioning, the fuzes shall be removed from the warheads and the electrical resistance shall be measured across the PIG leads. A fuze shall be classed defective if the resistance reading is more than 10,000 ohms. The meter used shall be current limited to 20 microamperes. The fuzes shall then be conditioned at the applicable temperature for a minimum of two (2) hours. All fuzes shall be subjected to the spin environment (see 3.5.6) within five (5) minutes after removal from the temperature conditioning chamber. Fuzes which are scheduled to be functioned by the IMP shall be returned to the temperature conditioning chamber for reconditioning prior to functioning. If data is lost because of an equipment failure during lot acceptance testing, another fuze from the same group of the lot shall be selected and subjected to the same conditioning as the unit being replaced. Fuzes shall be armed by application of a longitudinal acceleration of 33g's and then shall be explosively functioned in accordance with the applicable schedule. Fuzes scheduled for functioning by the Impact Switch (IMP) shall satisfy the requirements of 3.3.5. Fuzes scheduled for functioning by Self-Destruct (SD) shall satisfy the requirements of 3.3.6. A fuze shall be classed defective if it fails to satisfy any of the applicable requirements.

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NOTE: During lot acceptance testing only, the boost switch shall be monitored for proper closure. If the boost switch does not close at 33g's, the unit shall be removed from the centrifuge and tested on the impact machine with a longitudinal acceleration of 33g's and power applied for a minimum of 0.006 seconds. After the piston actuator fires and the timer runs down, the unit will be considered acceptable. However, another fuze must be selected from the same group of the lot to take the place in the lot acceptance test schedule of the unit functioned by the above procedure.

4.4.20 Environmental Testing (TM Fuze). - The specified number of samples shall be assembled to simulated warheads and then subjected to the required environmental conditions in the order shown for the specified environmental group (I, II or III). After completion of environmental conditioning, the fuzes shall be removed from the warheads and conditioned at the applicable temperatures for a minimum of two (2) hours. All fuzes shall be subjected to the spin environment (see 3.5.6) within five (5) minutes after removal from the temperature conditioning chamber. Fuzes which are scheduled to be functioned by the IMP shall be returned to the temperature conditioning chamber for reconditioning prior to functioning. If data is lost because of an equipment failure during lot acceptance testing, another fuze from the same group of the lot shall be selected and subjected to the same conditioning as the unit being replaced. Fuzes shall be armed by application of a longitudinal acceleration of 33g's and shall then be functioned by either the impact switch (see 3.3.5) or by self-destruct (see 3.3.6) as required by the test schedule. The arm signal and the functioning signal shall be monitored to assure a proper TM signal in accordance with 4.4.6. A fuze shall be classed defective if it fails to satisfy any of the applicable requirements.

NOTE: During lot acceptance testing only, the boost switch shall be monitored for proper closure. If the boost switch does not close at 33g's, the unit shall be removed from the centrifuge and tested on the impact machine with a longitudinal acceleration of 33g's and power applied for a minimum of 0.006 seconds. If the piston actuator fires and the timer runs down, the unit will be considered acceptable. However, another fuze must be selected from the same group of the lot to take the place in the lot acceptance test schedule of the unit functioned by the above procedure.

5. PREPARATION FOR DELIVERY

5.1 Packing and Marking

5.1.1 Level C. - The fuzes shall be packed in fiberboard cartons with fiberboard partitions.

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5.1.2 Packing. - Overpack the fiberboard cartons into a wooden box in accordance with MIL-B-2427, Type I, Class I, Grade C. Boxes will be closed and strapped per instructions contained in the Appendix of MIL-B-2427. Gross weight of the box shall not exceed 150 pounds maximum.

5.1.3 Marking. - Exterior boxes shall be marked Class C - Combination Fuzes - Handle Carefully - For Cargo Aircraft Transport Only.

5.1.4 Alternative Packing. - Overpack the fiberboard cartons into a wirebound box in accordance with MIL-B-46506, Type I, Class 2, Grade B. Packing and strapping shall be as found in the Appendix of MIL-B-46506. The same weight limitations apply as in 5.1.2 and marking as in 5.1.3.

6. NOTES

6.1 Ordering Data. - Procurement documents shall specify the following:

a. Title, number and date of this specification.
b. Ammunition data cards shall be prepared for each lot in accordance with MIL-STD-1167. The data cards shall contain the lot number and date of loading of the following explosive items: Piston Actuator, Detonator, MMS 1.0 and the Lead Cup Assembly; with each of these explosive items listed for each particular fuze in the lot.

6.2 Inspection Code Numbers. - The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government. All submitted designs shall contain a reference to the applicable five digit code number contained in Section 4 of this specification and the appropriate component or assembly drawing number and revision letter to which the specific design applies. These code numbers are used to correlate the characteristics cited on Equipment Lists with the inspections listed in this specification. They should also be cited as references on drawings of equipment designs submitted by the contractor (see 6.3.2)

6.3 Inspection Equipment Designs. - Inspection equipment designs are of two types - Government Special Inspection Equipment (SIE) designs and contractor designs. SIE designs are designated by drawing numbers on the Equipment Lists referenced on the Equipment Tabulation. Design responsibility for all other inspection equipment is assigned to the contractor. Equipment to be designed by the contractor is designated "CD". The contractor need not furnish any design when a complete Government SIE design is provided. Unless otherwise specified, however, the contractor may submit alternate or modified contractor designs of SIE in accordance with 6.3.2 and 6.3.3 should he elect to do so.

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6.3.1 SIE Designs. - SIE designs may consist of any of the following:

a. Detailed drawings which completely depict all information necessary for the fabrication and use of the item of inspection equipment.

b. A source control drawing or a specification control drawing as defined in MIL-STD-100.

c. An envelope drawing, as defined in MIL-STD-100, which establishes the criteria which a detail design shall meet. When envelope drawings are specified, the contractor shall prepare designs which comply with the criteria therein.

6.3.2 Contractor Designs. - Contractor designs are required for all inspection equipment for which SIE designs are not specified and may include commercial equipment which the contractor proposes to use. (Commercial equipment is defined as unmodified equipment which is cataloged and available for purchase by the general public.) Contractor designs shall include appropriate operating instructions, calibration procedures and maintenance procedures. Commercial equipment shall be fully described by catalog listings or other means which provide sufficient information to permit identification and evaluation by the Government and may include illustrations and engineering data. Designs shall be prepared for any special fixture(s) required to be used with commercial equipment or with SIE designs if not otherwise covered thereby (see 6.3.1.c). Designs shall be of the category and form (per MIL-D-1000) specified in the Contract Data Requirements Lists (DD Form 1423). The contractor is referred to MIL-HDBK-204, "Inspection Equipment Design" for guidance. The specification number and the applicable five-digit defect code number (or other specific identifying information) from Section 4 of this specification shall be referenced on each contractor design, together with the component or assembly drawing number and revision letter to which the specific design applies.

6.3.3 Submission of Designs for Approval. - Contractor designs shall be approved by the Government prior to fabricating or procuring the equipment. Designs shall be submitted for approval to Picatinny Arsenal, ATTN: SMUPA-QA-A-F, Dover, New Jersey 07801 in accordance with the stipulations, time frame and distribution specified in the Contract Data Requirements List (DD Form 1423) or in the contract. Partial submission of inspection equipment designs is permissible and encouraged. However, the completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will normally be accomplished within one month after receipt.

6.4 Submission of First Article Samples. - Instructions as to the location for the evaluation of the first article shall be obtained from the Contracting Officer. Upon receipt of such request, the Contracting Officer shall advise Picatinny Arsenal and instructions will be issued accordingly. All inquiries should be forwarded to: Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-QA-I, Dover, New Jersey 07801.

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6.5 Unarmed Fuze. - An unarmed fuze shall be defined as the lead 90° out of line from the detonator and the detonator electrically shorted.

6.6 Armed Fuze. - An armed fuze shall be defined as the lead in line with the detonator and the short removed from the detonator and connected to the firing circuit.

6.7 Copies of all lot acceptance data shall be sent to:

Commander
U.S. Army Missile Command
ATTN: AMSMI-QEA
Redstone Arsenal, Alabama 35807

Commander
U.S. Army Picatinny Arsenal
ATTN: SMUPA-QA-A-F/H. Bernstein
Dover, New Jersey 07801

If any failure occurs during lot acceptance testing, a failure analysis sheet shall be prepared on the failure and attached to the lot acceptance data. This sheet shall contain, as a minimum, analysis of failure and what corrective action is being taken to insure that the failure does not occur again.

NOTICE: When Government drawings, specifications or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished or in any way supplied said drawings, specifications or other data is not to be regarded by implication or otherwise in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

Custodian:
Army-MU

Preparing Activity:
Army-MU

Project Number: 1340-A-390

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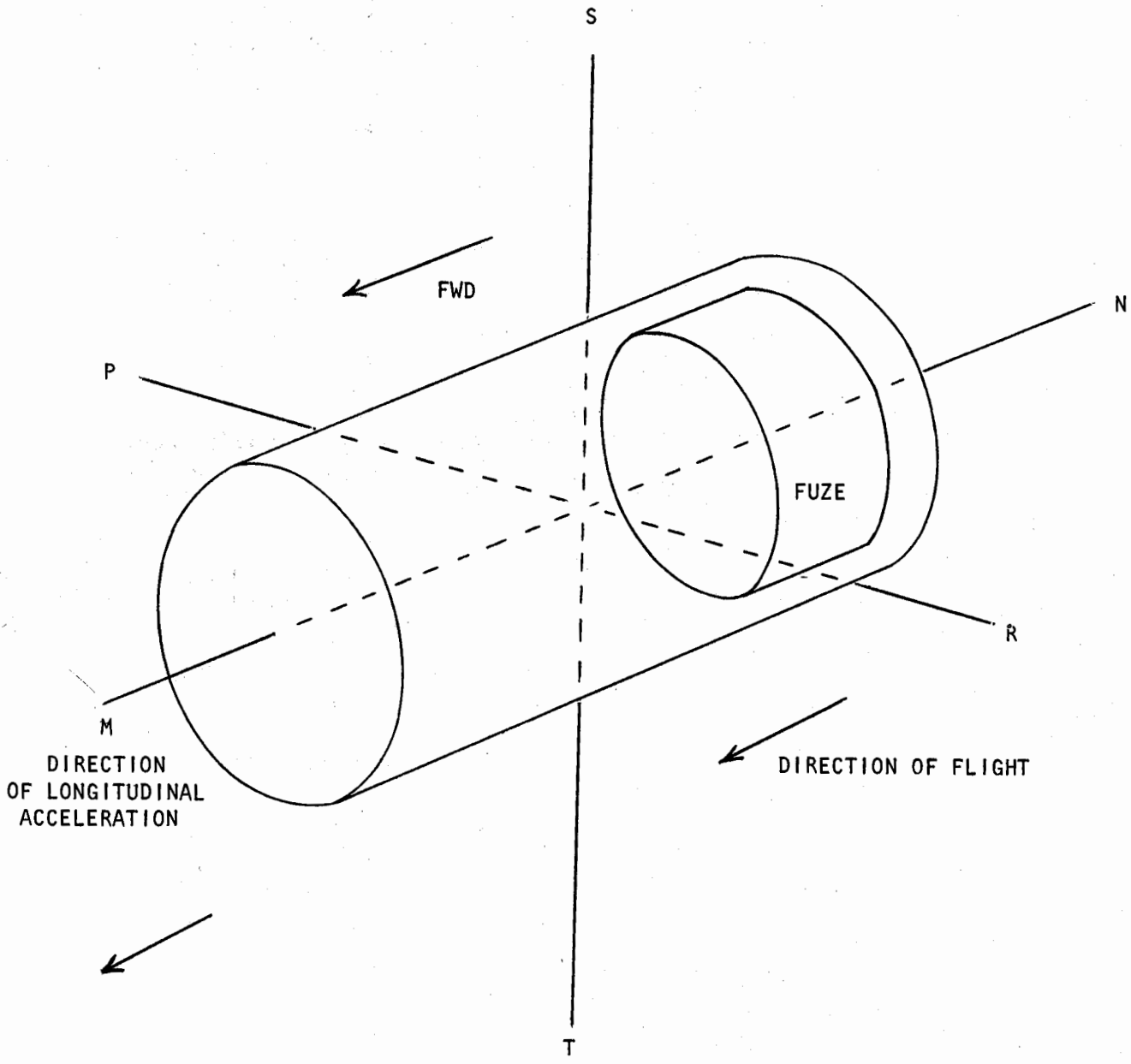


FIGURE 1. REFERENCE AXES

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